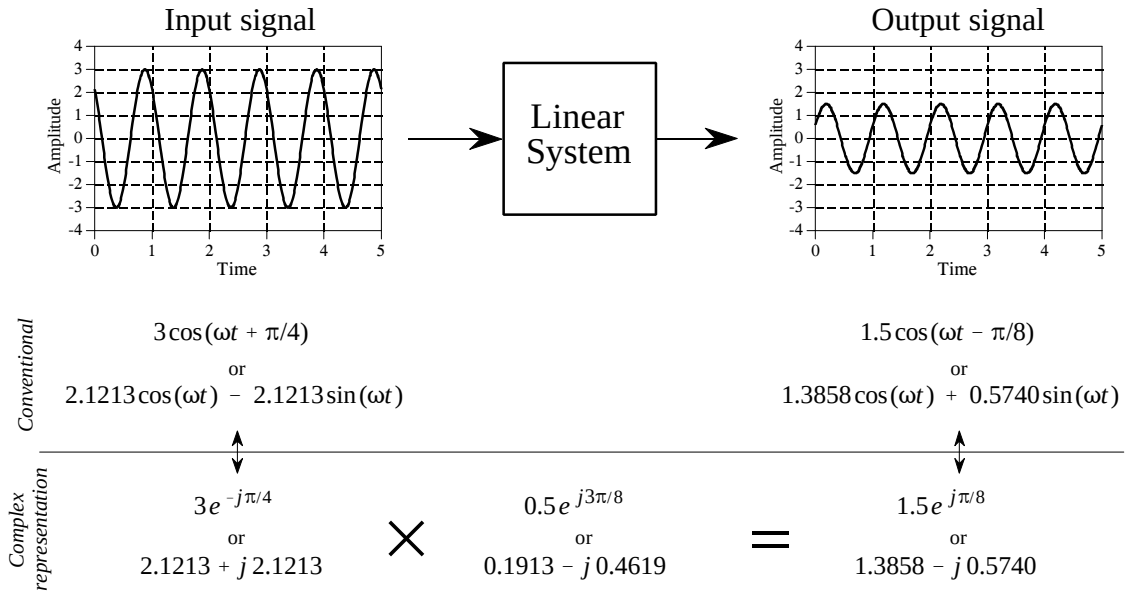

FIGURE 30-4
Sinusoids represented by complex numbers. Complex numbers are popular in DSP and electronics because they are a convenient way to represent and manipulate *sinusoids*. As shown in this example, sinusoidal input and output signals can be represented as complex numbers, expressed in either polar or rectangular form. In addition, the *change* that a linear system makes to a sinusoid can also be expressed as a complex number.

of the input and output (i.e., M_{out}/M_{in}). Likewise, the angle of the complex number is the *negative* of the difference between the input and output angles (i.e., $-[\phi_{out} - \phi_{in}]$). In the example used here, the system is described by the complex number, $0.5e^{j3\pi/8}$. In other words, the amplitude of the sinusoid is reduced by 0.5, while the phase angle is changed by $-3\pi/8$.

The complex number representing the system can be converted into rectangular form as: $0.1913 - j\,0.4619$, but we must be careful in interpreting what this means. It does *not* mean that a sine wave passing through the system is changed in amplitude by 0.1913, nor that a cosine wave is changed by -0.4619. In general, a pure sine or cosine wave entering a linear system is converted into a *mixture* of sine and cosine waves.

Fortunately, the complex math automatically keeps track of these cross-terms. When a sinusoid passes through a linear system, the complex numbers representing the input signal and the system are *multiplied*, producing the complex number representing the output. If any two of the complex numbers are known, the third can be found. The calculations can be carried out in either polar or rectangular form, as shown in Fig. 30-4.

In previous chapters we described how the Fourier transform decomposes a signal into cosine and sine waves. The amplitudes of the cosine waves are called the *real part*, while the amplitudes of the sine waves are called the *imaginary part*. We stressed that these amplitudes are ordinary numbers, and